Cybersecurity

Course Description

This is a full year course for students in 9th grade and above. It is designed to foster interest in Information Technology and networking careers. Students taking this course will develop the knowledge and skills required to identify and explain the basics of computing, IT infrastructure, applications, software development, database use, and security concepts. This will prepare students taking this course to take the CompTIA Tech+ (formerly ITF+) certification exam.

Course Objectives

- Identify and understand the purpose of computer components
- Understand how to implement the best IT and security practices
- Define and identify basics
- Compare and contrast different connector types
- Compare and contrast operating systems and their functions and features
- Define and identify basic security threats
- Perform proper steps to set up a basic workstation
- Prepare, take and pass the CompTIA Tech+ exam

Assessing Performance

Formative assessment includes worksheets, interactive activities, and quizzes for each lesson. Summative assessment includes projects and tests for each unit, with a final exam.

Equipment	Cost/Unit
Consumable Material, Reusable Material	$0
Classroom set of computers/Chromebooks	$0 if you already have, $500-$600 per device if you need to purchase
CompTIA Exam	$150

First Semester	
Unit 0: Introduction, Certification & Careers	Course introduction, explanation of certification exams and their importance in IT, career options
Unit 1: IT Concepts & Technology	Basics of computing, notational systems, units of measure, troubleshooting methodology
Unit 2: What is Cybersecurity?	Why cybersecurity matters and is needed
Unit 3: Applications & Operating Systems	Purpose of operating systems, OS components, purpose and proper use of software, web browsers, artificial intelligence
Unit 4: Risk, Adversity & Trust	Importance of trustworthiness in cyberspace
Unit 5: Cryptology	Codes
Unit 6: Software Development Concepts	Programming language categories, programming concepts
Unit 7: Hardware & Software Configuration	Internal computing components, storage types, peripheral devices, input/output devices and interfaces
Unit 8: Threats, Vulnerabilities & Attacks	Attacks and threat sources

Second Semester	
Unit 9: Data & Database Fundamentals	Value of data and information, database concepts, structures
Unit 10: Data Security	Data in cyberspace
Unit 11: Security	Fundamental security concepts and frameworks, best practices, encryption, wireless network installation
Unit 12: Malware & Digital Forensics	Malicious software and digital forensics
Unit 13: Introduction to Networking	Virtualization and cloud technologies, internet service types, networking concepts, authentication, authorization and accounting
Unit 14: States, Statelessness, Sovereignty & Cybersecurity	National Security Council, social values and ideologies
Unit 15: Final & Tech+ Certification Prep	Exam review and practice certification tests

CYBERSECURITY

1. Materials
   Internet access and 1-to-1 computer use daily.
   Chromebooks, Windows PCs, and Mac computers are acceptable for this course.
2. Required software, networking access, and access to LSU resources:
   - Students and teachers must have reliable access to the Louisiana Training Platform (LTP, http://louisianatrainingplatform.com)
   - Course lessons, instructional videos, interactive activities, assignments, quizzes, and other course resources are delivered through LTP.
   - District information technology departments should configure network filters, firewalls, and device security settings to allow student and teacher devices to access LTP and the resources required for course instruction. This includes linked websites, embedded videos, interactive activities, and online tools. Required resources should be tested before instruction begins to ensure that security settings do not block access or prevent activities from functioning as intended.
   - Students must be able to upload files (presentation, spreadsheet, document) through LTP when required by course assignments.
   - Whitelist (web-based activities)
     - Binary Game (https://learningcontent.cisco.com/games/binary/index.html)
     - US Cyber Range (https://www.uscyberrange.org/)
     - Packet.School (https://packet.school/en/sandbox)
     - Network Learning Platform | Packet.School
     - Cisco Packet Tracer | Cisco Packet Tracer
     - NetLab - Build Networks in Your Browser | Free Packet Tracer Alternative
     - Unit: Internet Simulator - CodeAI
     - CS4G Network Simulator
3. Requirements for IBC Alignment: **CompTIA Tech+ OR CIW Network Technology Associate (B) AND CIW Internet Business (B)***
   For schools who are pursuing the aligned IBC for this course
   a. Schools must have the appropriate testing software installed on the testing computers. For this credential the testing software is through CompTIA or CIW depending on the selected certification rouge.
      i. CompTIA: https://www.pearsonvue.com/us/en/comptia/onvue.html or https://www.pearsonvue.com/us/en/comptia/onvue.html
      ii. CIW: More information can be found here: https://ciwcertified.com/exam-requirements/
   b. Some industry based credential providers also provide practice tests softwares such a GMetrix which may be installed. Please check with individual schools if they plan to use a practice software.
   c. Schools must also typically register as an approved testing center prior to being allowed to proctor the exams.
   d. Teachers must have earned the IBC prior to students sitting for the exam. Once they have passed the credential, they need to be set up as a proctor in the ComptTia, PearsonVUE, or CIW

system. Please contact the appropriate vendor to be set up as a proctor depending on your selected IBC.

e. Teachers should report to the LSU Point of Contact their overall pass rates (i.e. 15/20 students passed)

4. Required teacher collaborations
   Teachers will communicate with LSU instructors via email. Teachers may be asked to share student work, course implementation feedback, or other instructional information with their designated LSU Pathways Point-of-Contact.
5. Required administration of course content, pre/post test, and research instruments
   Required course materials, assessments, activities, and instructional resources will be distributed through the Louisiana Training Platform or other LSU-approved systems identified by the course instructor.
6. Course Work
   Teachers must present the course material in sequence or as approved by collaboration with the LSU Pathway Point-of-Contact. Teachers are expected to deliver a minimum of 80% of the course material for course fidelity and for the opportunity for IBC attainment (where applicable).
7. Other
   As this is a project-based learning class, we strongly suggest that each section of the course be limited to a *maximum* of 20 students. The course is dependent on the teacher providing feedback and reviewing student code. The course requires that teachers have adequate time to interact with each student.